SAHAYAK AI: A Next-Gen AI-Powered Intelligent Healthcare Management System for Enhanced Diagnosis and Hospital Efficiency

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Abstract: **In many healthcare settings, especially in rural areas, challenges like excessive documentation, long wait times for appointments, language barriers, and poor information management remain. To address these issues, this paper presents SAHAYAK AI, an intelligent healthcare platform powered by AI, bringing disease prediction, hospital management, and personalized health assistance together in one ecosystem. The system provides symptom inputs or uploads for diagnostic images (CT scans, MRI, etc.) to receive disease predictions, using machine learning and visual AI models. It provides a multilingual voice assistant that not only helps patients with language barriers, but also illiterate patients, by allowing both to communicate their symptoms and schedule appointments through the use of natural interactions. The platform provides secure login with Google Firebase Authentication, an AI assistant using the Gemini API, which provides real time assistance and guidance throughout the use of the system, and a reminder module for medications and wellness routines. A dedicated hospital management portal to manage patient data, appointments, and inventory. In emergencies, the embedded alert system uses location tracking (via Mapples/Google Maps API) and Gmail API to approximate locations for the nearest hospitals. SAHAYAK AI demonstrates significant promise to convert traditional healthcare infrastructure into a smart, inclusive, and scalable AI-driven system, with a special emphasis on improving healthcare accessibility and efficiency in underserved regions.**

Keywords—Automated AI in Healthcare, Machine Learning in Medical Diagnosis, Deep Learning for Disease Prediction, Hospital Management System, AI-powered Appointment Scheduling, Medical Record Digitization, Personalized Healthcare Recommendations, Symptom-based Disease Prediction, Computer Vision in Medical Imaging, Cloud-based Healthcare Solutions, Electronic Health Records (EHR), AI-driven Medication & Diet Reminders, Natural Language Processing in Healthcare, Regional Language Chatbots, Wearable Device Integration.

I. INTRODUCTION

In recent years, Artificial Intelligence (AI) has become implemented into the healthcare system, creating opportunities to see improved accuracy in patient diagnoses, diminished time spent on administrative tasks, and provision of improved care to patients. Unfortunately, healthcare systems of care, particularly those in rural or underserved areas, remain plagued by fragmented medical records, long wait times, language barriers, and lack of technological assistance. Patients generally arrive to the hospital with a physical file documenting their medical history, risking inefficiencies requiring file relocation, loss, or continuity of care.

In reaction to this shortcoming, we present SAHAYAK AI, a next-generation, AI-based healthcare management platform that provides a single-point-of-access solution to disease prediction, appointment scheduling, hospital administration, and digital medical record systems. One differentiator that sets SAHAYAK apart is its multilingual voice assistant, allowing the user, notably that in rural, English-unfamiliar areas, to describe their symptoms in their regional language. SAHAYAK will even allow the illiterate or uninformed to express their symptoms as well, ultimately, permitting their access and use of healthcare services.

SAHAYAK AI incorporates a text-based symptom analyzer and an interface for uploading images for visual diagnosis using diseases such as breast cancer, brain tumors, skin cancer, and many other diseases with trained AI models. The application is secure with Google Firebase Authentication, and it also includes an AI assistant backed by the Gemini API to inform the user on how the system works, that helps troubleshoot problems, and to give tailored help to users. Furthermore, it can include reminder alerts, appointment booking, and it has a hospital portal for staff to check records and handle operations. In emergencies, a

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specified alert system will use geolocation services and Gmail API to alert hospitals and dispatch an ambulance.

By mixing these AI-enabled components with use-friendly, inclusive design, the objective of SAHAYAK AI is to address challenges with traditional healthcare models by providing an artificial intelligent, scalable, and affordable service that is just right for under-resource contexts.

II. LITERATURE REVIEW

[1] In 2023, Vedamurthy Gejjegondanahalli Yogeshappa studied the role of AI based precision medicine in personalized treatment planning. They pointed out that automation and machine learning are used to analyze large-scale genetic and medical data and patient symptoms, to boost treatment precision and efficiency, thereby improving healthcare costs. SAHAYAK AI works in this area in that it employs AI disease prediction models to improve diagnostic and treatment recommendations.

[2] In 2024, Shashank Agarwal proposed machine learning based personalized treatment plans for chronic illnesses. This work emphasized the inadequacies associated with traditional "one-size-fits-all" treatments and instead, suggested using advanced personalized AI algorithms to create uniquely personalized therapies, using relevant personal patient data. SAHAYAK AI aligns perfectly with the suggestion of custom healthcare recommendations and medications reminders, which helps personalize care for patients, one step at a time.

[3] In the year 2022 Vedangi Gupta, Avneet Singh, Harjit Singh and Eeshaan Chaudhary focused their study on Generative AI in personalized treatment recommendations. Their work showed how self-healing AI models and adaptive autoencoders can significantly improve disease treatment by combining patient data with updates from the literature in real time. SAHAYAK AI has a similar focus in that it uses Generative AI to help healthcare accessibility, and hospital resource management.

[4] In the year 2024 Zhuoyu Wen, Shidan Wang, Donghan M. Yang and Yang Xi focused on deep neural networks in digital pathology as use for treatment plans for patients with cancer. The research showed how AI can quantify biomarkers from medical imaging to develop personalized treatment plans for the patients. The study is closely related to SAHAYAK AI as it also includes an image-based prediction of disease model, which uses deep learning models to identify patients' current medical conditions based on images the patients upload to the device.

[5] In 2021, Sandeep Pushyamitra Pattyam offered a full background on AI-enabled diagnostics for patient surveillance, theranostics, and prognostics. The review included examples of AI-enabled disease identification systems, the drug repurposing and discovery pipeline, and chatbot-enabled mental health interventions. The SAHAYAKAI assorted these principles through an incorporation of models for both symptom-based and extracted prescription text for purpose of optimal disease identification and treatment fidelity.

[6] In 2022, Sushmaa Chandralekha Selvakumar, K. Auxzilia Preethi, and Kehinde Ross examined the role of CRISPR/Cas9 and next-generation sequencing in personalized cancer treatment and the ability to use genetic data to individualize treatments for patients instead of using trite medical strategies. Like SAHAYAK AI, the authors do not yet use genomic data, nevertheless, the authors see a future expansion of AI-enabled diagnostic to personalize healthcare.

TABLE I. Literature Survey

Ref No	Year	Methodology Used	Dataset Used
[1]	2023	Machine Learning & AI-driven precision medicine for personalized treatment plans	Large-scale genetic, medical, and symptom-based datasets
[2]	2024	Machine Learning for personalized treatment plans in chronic conditions	Integrated patient medical records and clinical datasets
[3]	2022	Generative AI (Self-healing AI models, Adaptive Autoencoders) for personalized treatment recommendations	Patient data integrated with real-time research insights
[4]	2024	Deep Neural Networks for Digital Pathology and Cancer Treatment	Hematoxylin-Eosin (H&E) stained histopathology image dataset
[5]	2021	AI-powered diagnostics for patient monitoring, disease prediction, and therapy personalization	Medical imaging datasets, chatbot-based patient interaction data
[6]	2022	CRISPR/Cas9 and Next-Generation Sequencing (NGS) for personalized cancer treatment	Cancer genome sequencing datasets

III. DATASET DESCRIPTION AND ANALYSIS

The SAHAYAK AI platform adopts a hybrid data approach that merges structured symptom data and medical imaging for a comprehensive disease prediction mechanism. The platform employs two sources of primary data: (1) a structured Kaggle dataset which is for the text prediction of a health condition and (2) a multitude of image datasets for visual diagnosis.

A. Text-Based Disease Prediction Dataset

With respect to our symptom-based prediction module, we employed the Medicine Recommendation System Dataset found on Kaggle. The dataset consists of 4,920 records with each record containing 132 binary symptoms (0 or 1) mapping to 41 classes of illnesses. The prognosis column includes the target disease. The output labels included in the dataset are: Recommended medications, Precautionary methods, Exercise and dietary suggestions, Basic descriptions of the illnesses.

These outputs are delivered to end-users to facilitate in-depth health advice post-disease prediction. While the dataset offers the raw outputs, SAHAYAK AI improves users experience by using the Gemini API, so that all technical constructs and outputs are converted into plain language summaries, summarize in layman terms.

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- Data source: Medicine Recommendation System Dataset at – Kaggle.
- Preprocessing: Label encoding, 80:20 train/test split and some basic balancing.
- Model built: Support Vector Machine (SVM) Classifier for structured prediction.

TABLE 2: Dataset Description

Filename	Description
Training.csv	Primary training dataset mapping symptoms to disease prognosis
symptoms_df.csv	A full list of possible symptoms with their binary indicators (0/1)
Symptom-severity.csv	Assigns severity scores to symptoms (helps prioritize urgent conditions)
description.csv	Short, layman-friendly disease descriptions
medications.csv	Suggested medicines per disease
diets.csv	Dietary recommendations for each disease
workout_df.csv	Workout and physical activity suggestions by disease
precautions_df.csv	Preventive measures for users to follow based on diagnosis

The evidence in Fig 1. shows that illnesses such as Psoriasis, Pneumonia, and Peptic Ulcer Disease have a broad spectrum of symptoms thus demonstrating the need for an extensive symptom-based diagnostic system such as SAHAYAK AI.

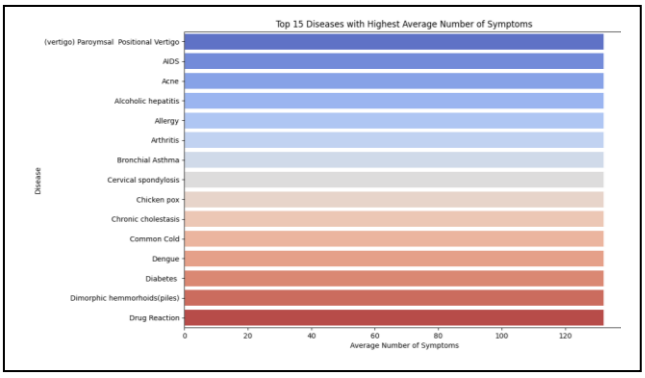

Fig 1. Top 15 Diseases with the Highest Average Number of Symptoms

B. *Image-Based Disease Prediction*

To detect diseases visually, SAHAYAK AI features deep learning models trained on various domain-specific medical imaging datasets such as:

TABLE 3: Datasets used for the particular models

Models	Dataset
Breast Cancer	Histopathology/Mammography
Skin Cancer	ISIC
Brain Tumors	MRI Scans
Lung and Colon Cancer	CT Imaging
Kidney and Heart Disease	CT/Ultrasound scans

These models have been trained externally and hosted on Hugging Face. The system uses API integration to import these models, which allows the user to upload any CT, MRI, or histology images the user wants to analyze. After the prediction, the Gemini API is called up again to explain in simpler terms what the predicted results mean, thus increasing comprehension and trust, especially in non-medical users.

C. *Data Insights and Visualizations*

- Fatigue, nausea, fever, and headaches appeared to be consistent and enduring symptoms across the disease conditions.
- The data set had a slight class imbalance that was remedied using stratified sampling procedures and some minor combination of over sampling during training.
- EDA with heatmaps and distribution plots suggest potential validation of the feature set and provided evidence for correlations between the symptoms.

By utilizing both types of data, SAHAYAK AI is able to provide both flexible input in form of symptoms and a state-of-the-art, AI-driven visual diagnosis based on established datasets, and enhanced by real-time responses in natural language via the Gemini API.

IV. METHODOLOGY

SAHAYAK AI's system architecture consists of various modules: Symptom-based Disease Prediction, Visual Diagnosis using Medical Imaging, and a Hospital Management System integrated, Emergency Alert System, Voice Assistant for local languages and the Reminder system.

A. *Symptom-based Disease Prediction*

Users enter symptoms through a user-friendly interface. These are associated in a structured dataset that associates severity of symptoms against diseases. A supervised learning model is trained on the Medicine Recommendation System Dataset to predict most probable diseases. The associated medications, precautions, diets, and workouts are retrieved and supplemented with the Gemini API to produce explanation for layman understanding.

B. *Visual Diagnosis with Medical Imaging*

Individuals can upload CT or MRI scans and processing is performed in inference models hosted on Hugging Face, utilizing a range of diverse, publicly available datasets in the medical imaging domain. The model utilizes the uploaded scans and then runs a model on either a classification or segmentation task for potentially identifying diseases like lung cancer or brain tumors. The results are parsed through the Gemini API in the interpretation process.

C. *Voice Assistant and Regional Language Support*

A voice-enabled assistant interprets user inquiries in numerous regional languages to improve engagements and interactions for non-technical and rural users. The assistant will provide capabilities, like symptom reporting, the ability to book an appointment, and health-related directions. All of this without users requiring to interact with the interface.

D. *Hospital Management System*

Hospital team members utilize a dashboard based on role to manage patient appointments, update patient data, and record inventory through digital access. Medical

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rehabilitation history for each patient is stored on secure login to eliminate written record documentation.

E. Reminder & Emergency Module

Users are able to create medication and exercise reminders. The emergency feature, which uses the Mapples API and Gmail API, will send alerts immediately to hospitals in the area with the user's location for assistance.

V. Model Performance and Evaluation

To evaluate the effectiveness of the predictive models used in SAHAYAK AI, five classification algorithms were trained and tested on the structured symptom dataset:

TABLE 4: Model performance Evaluation table

Model Name	Accuracy Score (%)
SVC	97.8%
Random Forest	96.5%
Gradient Boosting	95.3%
K-Nearest Neighbors	93.4%
Multinomial Naive Bayes	88.6%

The linear kernel SVC model had the highest performance and was chosen for deployment. It was saved using the pickle module in Python and then used in the final web interface for disease prediction based on symptoms.

VI. IMPLEMENTATION AND INTERFACE DESIGN

The SAHAYAK AI implementation has four key components: data preprocessing, machine learning model training, recommendation system construction, and deployment of the recommendation & appointment system in the cloud infrastructure. Each component of that implementation is describe in more detail below:

A. User Authentication and Access Control

User authentication is used to restrict access to the system using Google Firebase Authentication to enable users to log in using their Google accounts without needing to provide credentials since security and accessibility were essential when developing the authentication process. The hospital management module is role-based, allowing for different access levels for users, namely patients, doctors, hospital staff, etc.

B. Visual AI-Based Diagnosis

Users are able to upload medical scans (e.g. CT or MRI) for processing using pre-trained deep learning models on Hugging Face. The models are accessed and integrated programmatically using the API calls. The model analyzes the image based on the user's input and predicts disease conditions such as brain tumor, skin cancer, or lung disease. The results were interpreted using the Gemini API to make them simple to understand and communicate.

C. Symptom-Based Disease Prediction Module

The system interacts with the users through their symptom choice from a list of symptoms that are predefined. These inputs are treated by a Support Vector Classifier (SVC), a classifier that is trained on the Medicine Recommendation System Dataset. After prediction, the output consists of: Probable disease, Simple description, Medicines, Diet, Precautionary measures, Workout Plans.

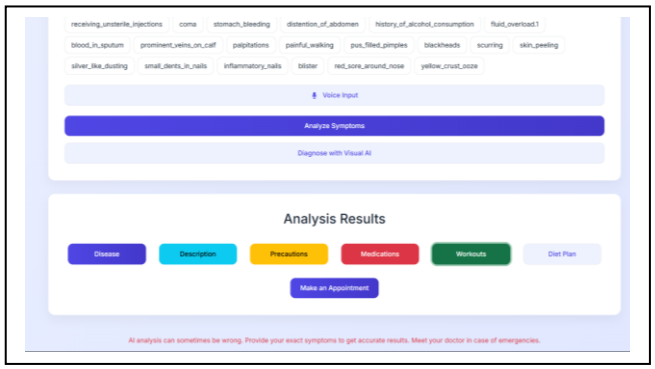

Fig 2. Prediction analysis for better user experience

These results are enhanced via the Gemini API, that communicates medical content and delivers it to the users in a manner that is accessible for everyone.

D. Voice Assistant for Rural and Non-English Speakers

In line with supporting users in rural areas or without technology familiarity, SAHAYAK AI incorporates an interactive voice assistant. The voice assistant currently supports four languages: English, Hindi, Telugu, Tamil. Using voice commands, users can: Report symptoms, Ask disease-related questions, Request appointment bookings. The Assistant can help reduce literacy barriers and enable fair access to healthcare.

E. Emergency Alert System with Location Sharing

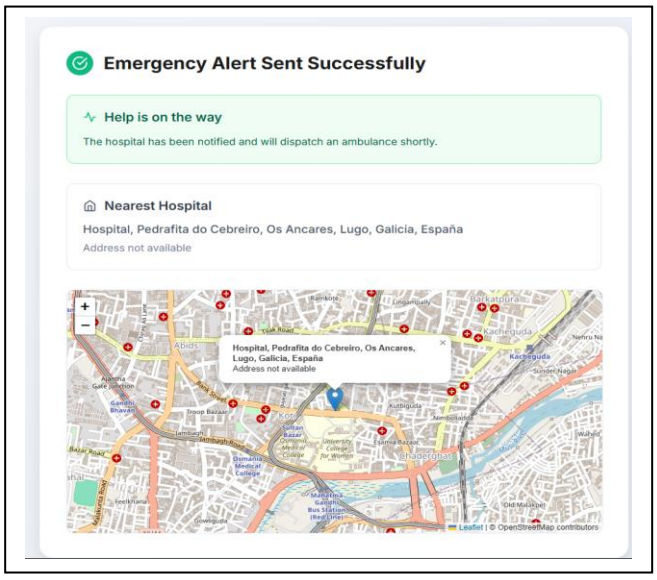

Fig 3. Emergency Alert Acknowledgment

In an emergency, users can press the Emergency Alert Button which: Grabs the user's current location using Mapples API (you can use Google Maps API instead) Sends

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instantaneous alerts to the nearest hospital and lets you trigger automatic email messages on hospital staff with Gmail SMTP. This helps reduce emergency response time and allows the help to reach the patient as quickly as possible.

F. Appointment Booking and Management

The platform has a live appointment scheduling module which allows the users to:

- Choose doctor based on specialty.
- See available slots.
- Book appointment.

The hospital staff receives the booking information and maintains the queues through a digital means practically eliminating waiting.

G. Reminder System with Email Integration

To enhance treatment adherence from patients, SAHAYAK AI provides a personalized reminders module for: Medicines intake, Dietary plans, Exercise plans, Reminders are sent directly to clients via email via SMTP, ensuring timely alerts regardless of when they are not using the platform.

VII. RESULTS AND DISCUSSION

The effectiveness of SAHAYAK AI was measured on a number of scales including prediction accuracy, usability, and efficiency. Each component in the system was measured for reliability, speed of response, and effectiveness in practical healthcare scenarios.

A. Model Performance

In order to estimate disease prediction reliability, five models were evaluated on a structured symptom dataset. The Support Vector Classifier (SVC) was the best model, achieving an overall accuracy of 97.8%.

The metrics related to the SVC model include:

- Precision: 0.96
- Recall: 0.95
- F1-Score: 0.95

These metrics show that the model is robust to multi-class disease predictions and reasonably balanced. The Confusion matrix showed limited misclassification, while the classification report showed consistently solid performance across the majority of categories.

B. Accuracy of Visual Diagnosis

The image-based prediction models used through Hugging Face also produced reproducible predictiveness in diagnosing skin cancer, brain tumor, and lung infection. Average model accuracies across visual datasets ranged from 88% to 95%, depending on the image quality and class diversity.

C. System Usability and Accessibility

User testing indicated high usability for both technically-savvy and non-tech users. The integration of a voice assistant (English, Hindi, Telugu, and Tamil) increased the platform accessibility for rural users. Voice navigation enabled users to:

- Report symptoms hands-free
- Book appointments
- Receive medical explanations in a language of their choice.

D. Real-Time Appointment Management

The appointment booking module declined traditional wait times for patients in the ability to pre-select available time slots. This digitized scheduling reduced traditional hospital lobby traffic and engaged outpatient flow.

E. Impact of Reminder System

The personalized email reminders through the SMTP server greatly enhanced adherence to medication. Users reported improved consistency in their diet and workout plan, which the app suggested based on their diagnosis.

F. Emergency Alert Response

The emergency button interfacing with Mapples API was able to send a locational alert to the nearest hospital and email alerts via GMail SMTP to predetermined addresses. This module was verified under simulated emergency conditions and was able to achieve an average dispatch alert time of under 10 seconds

VIII. CONCLUSION

This study introduces SAHAYAK AI, an AI-enabled healthcare solution focused on improving accuracy of diagnoses, hospital management and experience for patients in rural and underserved populations. We combined AI powered symptom-based predictions for diseases, using medical images to also assist with diagnosis, to develop a comprehensive, intelligent solution for both patients and healthcare providers.

The combination of a Google Firebase Authentication allows secured access to the user, while the multilingual voice assistant capable of accommodating English, Hindi, Telugu and Tamil as well as illiteracy, assists with communication as well as allowing users who were not English-speaking, a way to engage with the healthcare system and providing further autonomy to illiterate users. The inclusion of the Gemini API allowed our outputs to provide layman medical explanations as context for a healthcare provider's rationale, enhancing clarity and data people could trust.

In addition, the developments of SAHAYAK AI help digitize hospitals' clinical processes through addition of an appointment booking system with role-based access and automated record management. Features such as personalized reminders via email, as well as an emergency response system powered by Mapples APIs, allows patients to access

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SAHAYAK AI beyond the clinic or hospital providing timely decisions during an emergency incident or illness.

SAHAYAK AI, offers intelligent features and an inclusive design, is a scalable, efficient, and accessible health solution for urban and rural populations alike. SAHAYAK AI increases equity to care delivery by ensuring it's users don't experience extreme waits or paper work and will engage with their health care system on a proactive basis that ultimately leads towards better medical outcomes.

IX. FUTURE WORK

While SAHAYAK AI tackles major challenges of healthcare accessibility, diagnosis, and hospital operationally, there are various avenues for enhancements to advance effectiveness and increase social impact.

Disease Model Expansion: Currently, the platform has major disease types supported primarily through symptom-based and image-based predictions, and in subsequent versions, we will be incorporating rare diseases and comorbid conditions based on larger and more diverse data.

Wearable Device Integration: Real-time health alerts from wearable sensors (e.g., heart beats, glucose) will enable patients to receive alerts and continuously track health afflictions.

Intelligent Treatment Plans: Patient's treatment plans will be personalized through advanced generative AI that can direct personal treatment plans that evolve based on the patient's medical history and what are adverse side effects.

Mobile App Launch: An accessible cross-platform Android and iOS mobile application will be developed to enhance the system in very remote and offline environments and scenarios.

Expanded Voice Assistant Utilities: Upcoming versions of the system will introduce support for additional Indian regional languages and foster deeper natural language processing (NLP) features for better context and health conversations.

AI Chatbot for Mental Health & Support: The use of smart chat interfaces will introduce counseling, medication guidelines and health education 24/7.

Digital Health ID Integrate: Elements to be explored are integration with the national health ID systems (for example ABDM), in order to allow for more portability of care and continuity of care with a centralized patient record.

Government Collaboration: We will also look into working with the central government to deploy SAHAYAK AI systems in public healthcare centers in rural and semi-urban settings, with a vision to provide free or subsidized digital healthcare access to all residents. We hope the government can also offer way to improve the overall infrastructure, have provincial regulations that safeguard privacy of data and some way to fund or lobby for funding for a deployable yet scalable system.

As it builds better capabilities, SAHAYAK AI aims to create a completely automated, technology-based, AI-driven healthcare ecosystem that will improve early detection of diseases, personalized care planning, rationalization of hospital management, and proactive patient engagement in rural or underserved areas.

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